Local Income Taxes and Residential Sorting
Evidence from Pennsylvania LEHD Origin–Destination Data

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Motivation

- Tiebout (1956): citizens sort across jurisdictions according to their demands for local public goods.
- Strong evidence of tax-driven sorting at the state level (**??**).
- Little causal evidence on local income tax sorting within metro areas
- Two data problems make this hard. Local tax jurisdictions are sub-county, so IRS/ACS migration data are too coarse. Credit systems and weak enforcement complicate the effective tax burden.

This Paper

- Question: Do local income tax rate changes cause sorting?
- Setting: Pennsylvania, 2002–2019. Universal Earned Income Tax with continuous rate variation across ∼2,500 munis × ∼500 school districts.
- Data: LEHD LODES origin–destination, linking residence tract with workplace tract annually.
- Outcomes: `share_live_in` = (workers who live and work in city i) / (total workers in i), plus count outcomes (resident workers, total jobs).
- Exogeneous identification: Act 32 consolidated collection and made employer withholding mandatory.

Institutional Setting (Pennsylvania)

- Universal Earned Income Tax. About 2,500 munis and 500 school districts each set their own rate. Typical 0.5% muni + 0.5% school district.
- Before Act 32, collection was fragmented and withholding was inconsistent.
- Act 32 (effective 2012) consolidated collection and required employers to withhold the higher of the home and work rates.

How the PA EIT works (post-Act 32, excluding Philadelphia)

A worker pays the higher of the total home rate (muni resident plus school district) and the nonresident rate at the work location. The bill is split piecewise:

$$\left(\tau^{\text{home}},\ \tau^{\text{work}}\right) = \begin{cases} \left(\tau_j^{\text{res}} + \tau_j^{\text{SD}} - \tau_i^{\text{nr}},\ \tau_i^{\text{nr}}\right) & \text{if } \tau_j^{\text{res}} + \tau_j^{\text{SD}} > \tau_i^{\text{nr}} \\ \left(0,\ \tau_i^{\text{nr}}\right) & \text{if } \tau_j^{\text{res}} + \tau_j^{\text{SD}} \leq \tau_i^{\text{nr}} \end{cases}$$

- The work jurisdiction always collects its full nonresident rate.
- The home muni and school district then split whatever remains of the worker's bill. They get nothing if the work rate is higher.
- Which rate binds determines how the worker can respond. When the home rate binds, the worker can lower the bill by moving home. When the work rate binds, the worker can lower it only by changing jobs.

Worked example

Worker 1 lives in city B and works in city A. Home muni rate is 1.0%, home school district is 0.5%, so the total home rate is 1.5%. The work-location nonresident rate is 0.8%.

- Worker 1 pays the higher of the two rates, so 1.5%. Home collects $1.5 - 0.8 = 0.7\%$; work collects 0.8%.
- This is a home-binds case. To lower the bill, worker 1 would have to move out of B. Changing jobs would not help because the work rate is already lower.

Philadelphia: a separate regime

- Philadelphia is exempt from Act 32. Its Wage Tax operates under the Sterling Act of 1932, administered by the City Department of Revenue.
- Philly residents pay the Philly resident rate regardless of where they work. There is no max rule and no offset.
- Nonresidents working in Philly pay the Philly nonresident rate, which goes entirely to Philly.
- In my analysis I keep Philly in the main sample because it is part of PA's labor market, but exclude it in robustness checks because its reporting and dynamics differ from Act-32 munis.

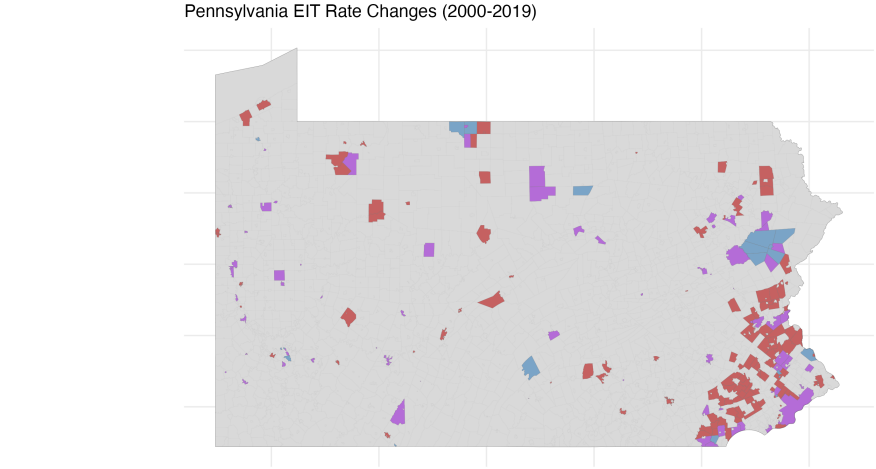
Data Generating Process

1. Pull Census tract LODES OD flows (LEHD, 2002–2019).
2. Scrape PA rates from DCED MunStats: muni resident, muni nonresident, and school district EIT for each muni-year.
3. Scrape PA municipal Annual Financial Reports (DCED) for spending on police, fire, streets, sewer, parks, comm dev, debt, and state aid.
4. Crosswalk DCED munis to Census (99.6% match) and aggregate tract-pair flows to muni-pair-year.

Summary Statistics (Pennsylvania, 2002–2019)

Variable	Mean	Median	SD	Max
share_live_in	0.103	0.088	0.076	0.657
Muni resident EIT (%)	0.532	0.500	0.175	4.331
Muni nonres EIT (%)	0.891	1.000	0.235	3.820
SD EIT (%)	0.617	0.500	0.289	2.050
Total jobs	2,438	565	14,502	629,698
Resident workers	450	50	8,001	391,047
Muni × year	35,743 (2,275 munis)			

Treatment Geography: PA Rate Changes



Red = increased; blue = decreased; purple = both; grey = unchanged.

Three Channels of Adjustment

When muni A raises its rate, residents of A can:

1. Move residence to a lower-rate muni, keep job in A. Saves at least the home SD rate at A.
2. Move both residence and job to a lower-rate muni. Largest savings.
3. New entrants pick a home outside A for a given job.

All three reduce `share_live_in` for A.

Outcomes

Unit i = PA municipality, t = year, 2002–2019.

- $\texttt{share_live_in}_{it} = \dfrac{\text{live + work in } i}{\text{total workers in } i}$
- $\log(\text{Resident workers}_{it})$ = workers who live in muni i, any work location.
- $\log(\text{Total jobs}_{it})$ = workers employed in muni i, any home location.

Q2: Do rate increases cause sorting?

Two specs

$$\text{Total stack:} \quad Y_{it} = \alpha_i + \gamma_t + \beta\, \tau_{it}^{\text{home}} + \mathbf{X}_{it}'\boldsymbol{\phi} + \varepsilon_{it}$$

$$\text{Decomposition:} \quad Y_{it} = \alpha_i + \gamma_t + \beta^{m} \tau_{it}^{\text{res,muni}} + \beta^{sd} \tau_{it}^{\text{SD}} + \mathbf{X}_{it}'\boldsymbol{\phi} + \varepsilon_{it}$$

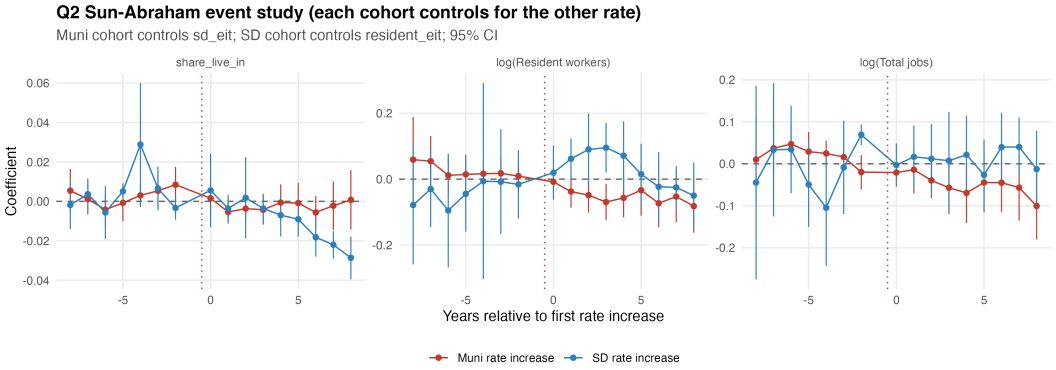
- Treated = munis that ever raised their muni or SD rate (within-city rate change drives identification).
- Control = munis whose rates stayed flat over the panel.
- $\mathbf{X}_{it}$ = log muni spending on police, streets, debt, state aid. Y_{it} = `share_live_in` or log count outcomes.

Q2 Answer: muni rate drives sorting; SD null

	share_live_in	log(Resident)	log(Total Jobs)
Total stack			
τ^{home}	-0.010^{**}	-0.137^{***}	-0.096^{**}
	(0.004)	(0.043)	(0.038)
Decomposition			
$\tau^{\text{res,muni}}$	-0.012^{**}	-0.201^{***}	-0.128^{***}
	(0.006)	(0.041)	(0.038)
τ^{SD}	-0.005	$+0.029$	-0.013
	(0.006)	(0.114)	(0.093)
R^2	0.909	0.975	0.976
N	12,891	12,781	12,891

- Controls: log muni spending (police, streets, debt, state aid) + revenue shares (EIT, real estate).
- A 1 pp muni rate hike lowers share_live_in by 1.2 share-pp and resident workers by 20%.

Q2 Event Study: muni vs SD decomposed



Q2 DCDH: CONTINUOUS TREATMENT, STAGGERED DESIGN

de Chaisemartin & D'Haultfœuille (2024) continuous estimator handles staggered treatment + continuous dose without TWFE's negative-weighting problem.

$$Y_{it} = \alpha_i + \gamma_t + \text{ACR} \cdot \tau_{it}^{\text{res,muni}} + \mathbf{X}_{it}'\boldsymbol{\phi} + \varepsilon_{it}$$

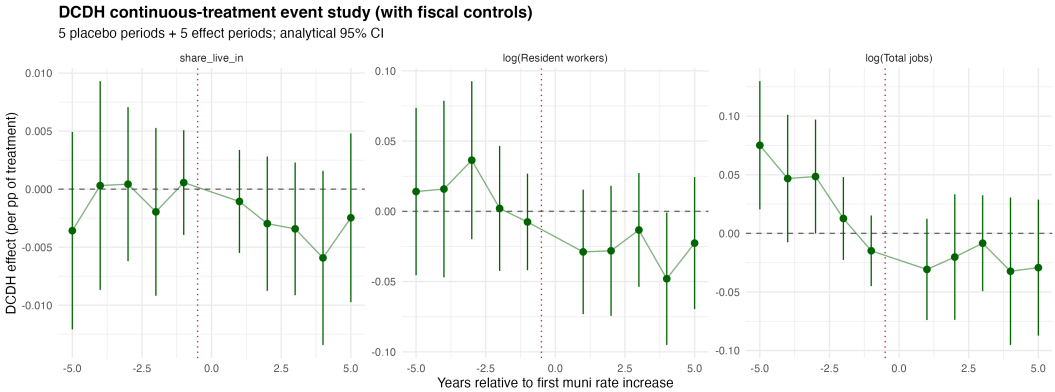
- Treated = munis ever raising muni rate.
- Control = munis whose rates stayed flat over the panel.
- Dose = muni resident rate (continuous).
- Effects up to $\ell = 5$; placebos up to $\ell = -5$.
- $\mathbf{X}_{it}$ = log muni spending on police, streets, debt, state aid. Y_{it} = `share_live_in` or log count outcomes.

Q2 DCDH Answer: average causal response per pp

	share_live_in	log(Resident)	log(Total Jobs)
ACR per pp	−0.009	−0.078	−0.067
	(0.006)	(0.047)	(0.055)
N (city-yrs)	12,926	12,926	12,926
Munis	948	948	948

- **Note: DCDH not yet rerun on the block-aggregated panel** (computationally intense). Numbers above are from prior tract-centroid panel; expected to shift modestly toward the new TWFE muni point estimates (-0.012^{**}, -0.201^{***}, -0.128^{***}).
- DCDH addresses the staggered-DiD critique: no negative weights from already-treated comparison groups.

Q2 DCDH Event Study



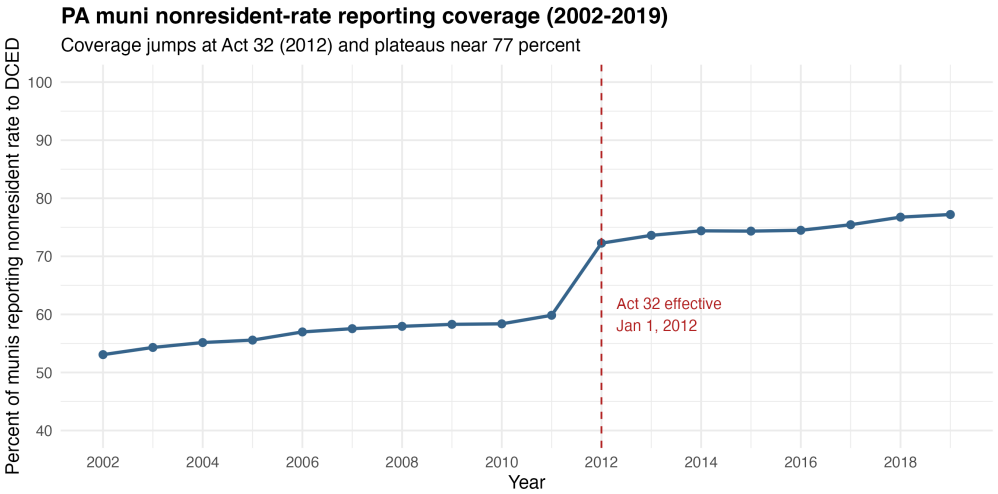
Mechanism: Is it enforcement? (Act 32)

Unit i = PA municipality; t = year. Outcome $\texttt{share_live_in}_{it}$. Structural break on rate effect:

$$Y_{it} = \alpha_i + \gamma_t + \beta^{\text{pre}} \tau_{it} \cdot \mathbf{1}[t < 2012] + \beta^{\text{post}} \tau_{it} \cdot \mathbf{1}[t \geq 2012] + \varepsilon_{it}$$

- Pre-2012: fragmented collection, weak withholding; DCED's nonresident rate registry was incomplete (only ∼55% of munis reported nonres rates; jumped to ∼72% in 2012).
- Post-2012: centralized, mandatory employer withholding at $\max(\tau^{\text{home}}, \tau^{\text{nr,work}})$.
- Run break for both rates separately to see which channel is enforcement-bound.

Nonresident rate reporting coverage



Do the rate increases we exploit actually bind?

For self-pair (worker lives and works in muni X): home stack $= \tau_X^{\text{res}} + \tau_X^{\text{SD}}$ vs. work nonres rate τ_X^{nr}.

- Across all PA muni-years post-Act-32 (15,423): home binds 41%, work binds 59%. Mean wedge $\tau^{\text{home}} - \tau^{\text{nr}} = +0.27$ pp.
- Of muni-resident rate increases 2012–2019 (57 events): **84% (48 of 57)** occur where home binds at the muni ⇒ the rate increase *moves the bill* for residents working locally.
- The remaining 16% (9 of 57) occur where work already binds and stays binding ⇒ the rate hike is irrelevant to in-muni workers' bills.

Cross-pair within 80 km, post-Act-32 (1.83M pair-years):

- Work binds (home stack $\leq$ work nonres): 61.6%.
- Home binds (home stack $>$ work nonres): 38.4%.

Implication: most of the resident-rate increases we exploit are economically meaningful for the in-muni worker population. The piecewise binding test below sharpens this: separates extensive switches in binding side from intensive within-channel rate changes.

Act 32 Answer: Q2 results split pre/post 2012

	share_live_in	log(Resident)	log(Total Jobs)
Decomposition			
$\tau^{\text{res,muni}} \times$ Pre	-0.008	-0.139^{***}	-0.073
$\tau^{\text{res,muni}} \times$ Post	-0.013^{**}	-0.209^{***}	-0.132^{***}
$\tau^{\text{SD}} \times$ Pre	-0.007	$+0.032$	$+0.027$
$\tau^{\text{SD}} \times$ Post	-0.003	$+0.026$	-0.038
N	12,891	12,781	12,891

- Controls: log spending (police, streets, debt, state aid) + revenue shares (EIT, real estate).
- Pre-Act-32 muni coefficient on share_live_in is null; post-Act-32 it is significant. Magnitudes on log(Resident) more than *double* from -0.139 pre to -0.209 post.
- SD coefficient is null in both periods — the prior $+0.147^{**}$ pre / $+0.170^{**}$ post pattern was an artifact of tract-centroid job misattribution.

Q3 Sample: clean treatment vs control groups

	Never-changed		Inc-only		Dec-only	
	Mean	SD	Mean	SD	Mean	SD
share_live_in	0.102	0.067	0.132	0.080	0.104	0.061
Muni resident EIT (%)	0.514	0.102	0.822	0.448	0.617	0.253
SD EIT (%)	0.590	0.253	0.556	0.214	0.483	0.090
Total EIT (%)	1.104	0.257	1.378	0.498	1.101	0.254
Total jobs	2,943	7,419	8,080	10,589	3,436	3,059
Resident workers	328	1,660	1,195	2,719	352	382
Munis	1,110		90		15	
Muni-years	11,076		1,126		178	

- Never-changed = munis with no rate change 2002–2019. Inc-only = ever raised, never decreased. Dec-only = ever decreased, never raised.
- Flip-floppers (raised AND decreased at some point) dropped.

Q3: Are rate changes symmetric? (clean groups)

Single regression: continuous muni rate interacted with group; SD rate enters as a single coefficient (no group interaction):

$$Y_{it} = \alpha_i + \gamma_t + \beta_g^m \tau_{it}^{\text{res,muni}} \cdot G_i + \beta^{sd} \tau_{it}^{\text{SD}} + \mathbf{X}_{it}'\boldsymbol{\phi} + \varepsilon_{it}$$

where $G_i \in \{\text{nc, inc, dec}\}$ defined by muni-rate event.

- Treated: inc-only munis (90) and dec-only munis (15).
- Control: never-changed munis (1,110).
- Flip-floppers dropped.
- Continuous rate (per pp), not binary cohort.
- $\mathbf{X}_{it}$ = log muni spending (police, streets, debt, state aid) + revenue shares (EIT, real estate, intergov).
- $\tau^{\text{res,muni}}$: nc dropped (no within-variation by definition).

Q3 Answer: Inc effect robust; SD null

	share_live_in	log(Resident)	log(Total Jobs)
$\tau^{\text{res,muni}} \times$ Inc	-0.016^{**}	-0.221^{***}	-0.151^{***}
	(0.007)	(0.038)	(0.040)
$\tau^{\text{res,muni}} \times$ Dec	$+0.032^{**}$	-0.295	-0.067
	(0.016)	(0.657)	(0.114)
τ^{SD}	-0.004	$+0.024$	-0.018
	(0.006)	(0.122)	(0.099)
N	12,298	12,188	12,298

- Controls: log expenditures (police, streets, debt, state aid) + revenue shares (EIT, real estate).
- Muni rate hits inc-only hard: -1.6 share-pp per pp; -22% on resident workers; -15% on total jobs.
- Dec-only point estimates noisy (only 15 munis). $+0.032^{**}$ on share_live_in is a small-sample composition effect, not a substantive sign reversal.
- SD coefficient is null — the prior $+0.139^{**}$ on log(Resident) was an artifact of

Q3 Full Results (all coefficients)

	share_live_in	log(Resident)	log(Total Jobs)
$\tau^{\text{res,muni}} \times$ Inc	-0.016^{**} (0.007)	-0.221^{***} (0.038)	-0.151^{***} (0.039)
$\tau^{\text{res,muni}} \times$ Dec	$+0.032^{**}$ (0.016)	-0.296 (0.657)	-0.067 (0.114)
τ^{SD}	-0.004 (0.006)	$+0.024$ (0.122)	-0.018 (0.099)
log_police	-0.000 (0.001)	$+0.004$ (0.012)	$+0.010$ (0.012)
log_streets	-0.001 (0.001)	$+0.010$ (0.010)	$+0.018^{**}$ (0.009)
log_debt	-0.000 (0.001)	-0.005 (0.005)	$+0.001$ (0.005)
log_state_aid	$+0.001$ (0.001)	$+0.011$ (0.010)	-0.005 (0.009)
share_eit	$+0.005$ (0.010)	-0.046 (0.118)	-0.105 (0.114)
share_re	-0.001 (0.008)	-0.076 (0.092)	-0.074 (0.088)
N	12,298	12,188	12,298

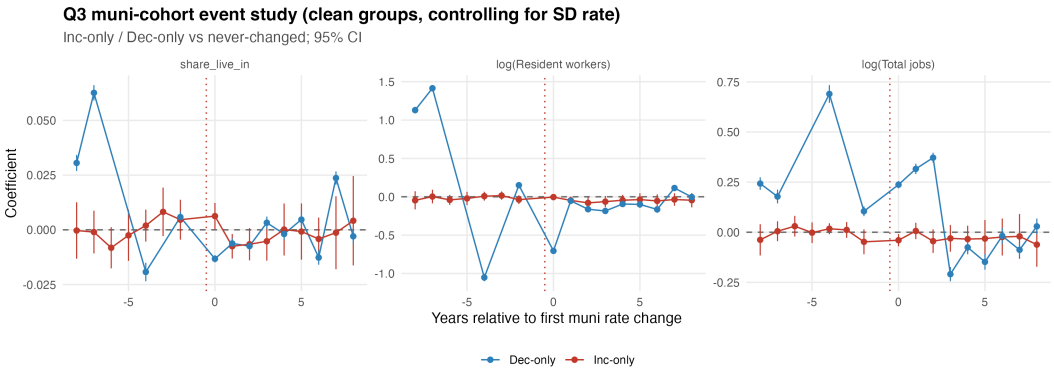
Cluster-robust SEs in parentheses. City + year FE.

Q3 Bonus: How SD rate changes track muni-event groups

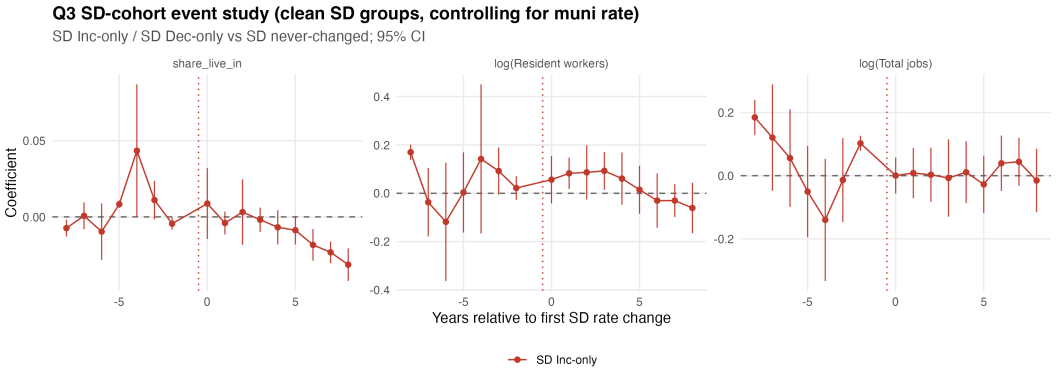
Muni group \ SD group	SD dec	SD flip	SD inc-only	SD nc
Muni dec-only (24 munis)	8%	0%	**50%**	42%
Muni inc-only (97 munis)	5%	0%	17%	78%
Muni nc (1,264 munis)	0%	0.4%	11%	89%

- Cities that cut their muni rate are *half* likely to be in an SD that raised its rate (50% vs 11% baseline).
- Cities that raise muni rates and never-changers mostly have stable SD rates (78–89% SD nc).
- Suggests a fiscal substitution pattern: SD raises rate → muni cuts to manage total burden, OR muni cut creates room for SD to raise. Act 1 of 2006 (PA SD rate referenda) was designed partly around this trade-off.

Q3 Event Study: Muni-cohort (Inc/Dec vs Never-changed)



Q3 Event Study: SD-cohort (Inc/Dec vs Never-changed)



SD inc-only ≈ 165 munis; SD dec-only too thin to fit.

Q3 Robustness motivation: staggered + multi-event treatment

- Of 113 inc-only munis, 17 (≈15%) raise rates more than once; 1 raises 9 times.
- Pooled TWFE on continuous `resident_eit` can have *negative weights* under heterogeneous treatment effects (Goodman-Bacon 2021, de Chaisemartin & D'Haultfoeuille 2020).
- Even after restricting to single-event munis, staggered timing (different munis treated in different years) keeps the problem alive — need SA / DCDH / stacked DiD.
- Four robustness checks below, each with both an aggregate coefficient and an event-study figure.

(1) Single-event TWFE: drop multi-event Inc munis

$$Y_{it} = \alpha_i + \gamma_t + \beta\, \tau_{it}^{\text{res,muni}} + \delta\, \tau_{it}^{\text{SD}} + \mathbf{X}_{it}'\boldsymbol{\phi} + \varepsilon_{it}$$

Sample: 1,110 NC + 68 Inc-only with exactly one rate increase (drops 22 multi-event Inc munis). $\mathbf{X}_{it}$: log spending + revenue shares. SEs clustered at muni.

	share_live_in	log(Resident)	log(Total Jobs)
Total stack			
τ^{home}	−0.004 (0.005)	−0.062 (0.072)	−0.077 (0.061)
Decomposition			
$\tau^{\text{res,muni}}$	−0.004 (0.007)	−0.162*** (0.035)	−0.144*** (0.047)
τ^{SD}	−0.004 (0.006)	+0.027 (0.122)	−0.020 (0.099)
log_police	−0.000 (0.001)	+0.005 (0.012)	+0.011 (0.012)
log_streets	−0.001 (0.001)	+0.011 (0.010)	+0.018** (0.009)
log_debt	−0.000 (0.001)	−0.005 (0.005)	+0.001 (0.005)
log_state_aid	+0.001 (0.001)	+0.011 (0.010)	−0.005 (0.009)
share_eit	+0.003 (0.011)	−0.052 (0.125)	−0.104 (0.121)
share_re	−0.001 (0.008)	−0.076 (0.092)	−0.074 (0.088)
N	11,828	11,719	11,828

Cluster-robust SEs by muni in parentheses. City + year FE.

(1) Single-event sample ES (Sun-Abraham, binary cohort)

$$Y_{it} = \alpha_i + \gamma_t + \sum_{\ell \neq -1} \mu_\ell \, \mathbf{1}\{t - c_i = \ell\} + \mathbf{X}_{it}'\boldsymbol{\phi} + \varepsilon_{it} \quad \text{(SA cohort-aggregated)}$$

SA aggregates the cohort × event-time interactions of **?** using cohort-share weights. Cohort c_i = year of muni i's first rate increase. **Note: SA needs binary staggered treatment, so the dynamic coefficients here are the average ATT in event-time, not the per-pp dose magnitude in the previous slide.**



(2) DCDH: CONTINUOUS, MULTI-EVENT, INC-ONLY + NC

de Chaisemartin & D'Haultfoeuille (2024). For each pair of consecutive years $(t-1, t)$ and each muni i, define the *instantaneous causal response*

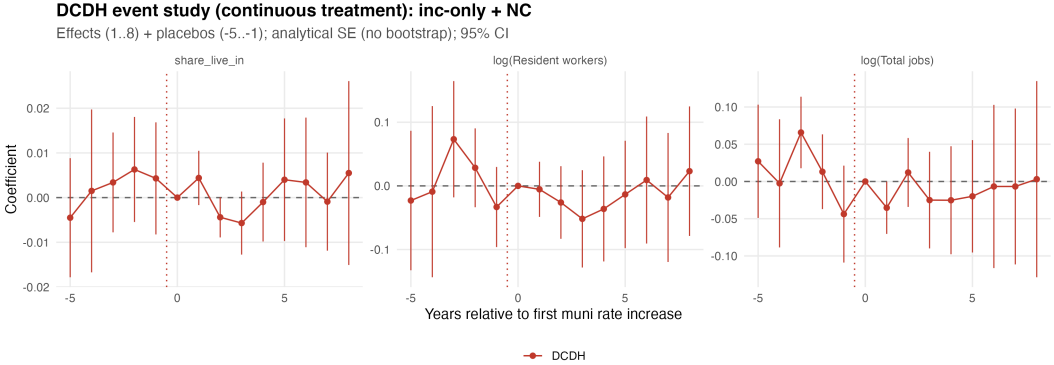
$$\delta_{i,t} = \frac{Y_{it} - Y_{i,t-1}}{\tau_{it} - \tau_{i,t-1}}$$

on switchers (munis with $\tau_{it} \neq \tau_{i,t-1}$). The estimator averages $\delta_{i,t}$ across switchers and event-times, using only never-treated munis as the difference-in-differences counterfactual; no negative weights even with continuous, staggered, multi-event treatment. ATE per pp:

	share_live_in	log(Resident)	log(Total Jobs)
Av. total effect	+0.002	−0.049	−0.043
	(−0.023, +0.026)	(−0.238, +0.139)	(−0.223, +0.137)
Effective N (cells)	10,853	10,750	10,853
Switchers	234	234	234

- ATE null but underpowered: only 36 switchers in t=1, falling to 21 by t=8.
- Analytical SEs are conservative under continuous treatment (package warns; bootstrap recommended for inference, ∼6–8 hr).
- Same controls as TWFE in the data setup; DCDH uses the residualized treatment after partialling controls.

(2) DCDH EVENT STUDY



Effects (years 1–8) + placebos (−5 to −1). Placebos not individually significant. Effect on log(Total jobs) at t=1 is −0.035 (95% CI −0.070, −0.000).

(3) Stacked DiD, binary post: 21 sub-experiments

For each cohort c (event year), construct a stack s: that treated muni plus all NC, $t \in [c-5, c+5]$. Concatenate stacks and estimate

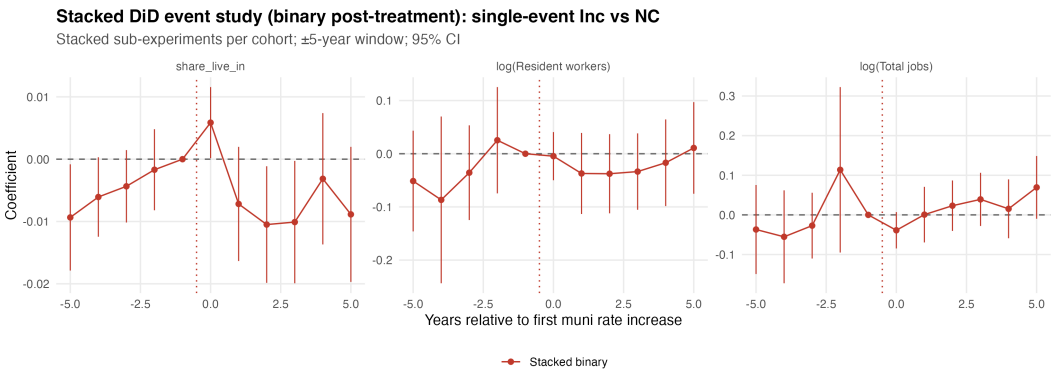
$$Y_{ist} = \alpha_{is} + \gamma_{ts} + \beta\,(\text{Treat}_i \cdot \text{Post}_{st}) + \mathbf{X}'_{it}\boldsymbol{\phi} + \varepsilon_{ist}$$

with stack×muni and stack×year FE. Only NC munis as controls ⇒ no already-treated comparison; SEs clustered at the original muni.

	share_live_in	log(Resident)	log(Total Jobs)
`treat_post`	−0.002 (0.003)	−0.000 (0.037)	+0.008 (0.029)
log_police, log_streets, log_debt, log_state_aid, share_eit, share_re	*(included; coefficients suppressed)*		
N (stack-muni-year)	95,287	94,317	95,287

- ATT pools all post-treatment years at intensive treatment = 1.
- Null because the intensity of the rate change is collapsed out — a 0.1pp hike and a 0.5pp hike both contribute equally.

(3) Stacked DiD binary event study



(4) Stacked DiD, continuous $\Delta\tau$: dose-response per pp

Same stacks as (3). Within each stack, define $\Delta\tau_{ist} = \tau_{it}^{\text{res,muni}} - \bar{\tau}_{is}^{\text{pre}}$ (rate deviation from the muni's own pre-treatment baseline within stack s). Then

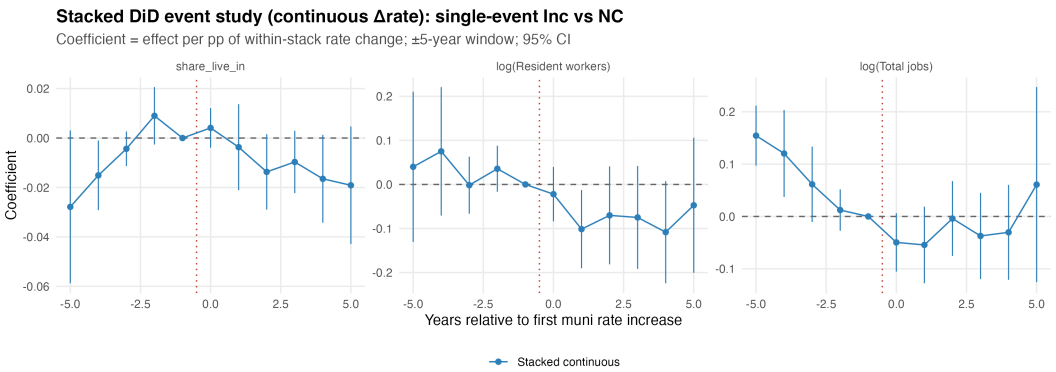
$$Y_{ist} = \alpha_{is} + \gamma_{ts} + \beta\,\Delta\tau_{ist} + \mathbf{X}_{it}'\boldsymbol{\phi} + \varepsilon_{ist}$$

$\hat{\beta}$ has a per-pp interpretation. The event-study version interacts a time-invariant per-unit dose with $\mathbf{1}\{t - c_i = \ell\}$.

	share_live_in	log(Resident)	log(Total Jobs)
$\Delta\tau$ (per pp)	−0.005 (0.005)	−0.091** (0.036)	−0.072** (0.031)
log_police, log_streets, log_debt, log_state_aid, share_eit, share_re	*(included; coefficients suppressed)*		
N (stack-muni-year)	77,182	76,528	77,182

- **Recommended headline magnitude.** 1 pp muni rate increase ⇒ −9% resident workers, −7% total jobs.
- Roughly half the magnitude of Q2 TWFE pooled (−20%, −13%) — consistent with TWFE being inflated by negative weights under heterogeneous effects.
- No staggered-DiD critique: only NC munis as controls, no already-treated comparison.

(4) Stacked DiD continuous event study



(5) Piecewise binding model: setup

Pair-level (work A, home B, year t). Effective bill is

$$\tau^{\text{bill}}(A,B,t) \;=\; \max\!\big(\tau_B^{\text{home}},\; \tau_A^{\text{nr}}\big), \qquad \tau_B^{\text{home}} = \tau_B^{\text{res}} + \tau_B^{\text{SD}}$$

Year-over-year per-pair binding-state transition:

- **ext_to_home**: $t{-}1$ work-binds, t home-binds
- **ext_to_work**: $t{-}1$ home-binds, t work-binds
- **int_home**: home-binds in both years (intensive on home)
- **int_work**: work-binds in both years (intensive on work)

Decomposition of $\Delta\tau^{\text{bill}}$ by transition type:

$$\begin{aligned}\Delta \log \text{flow}_{ABt} = {} & \beta_{ih}\,\Delta\tau_t^{\text{bill}}\cdot\mathbf{1}\{\text{int_home}\} + \beta_{iw}\,\Delta\tau_t^{\text{bill}}\cdot\mathbf{1}\{\text{int_work}\} \\ & + \beta_{eh}\,\Delta\tau_t^{\text{bill}}\cdot\mathbf{1}\{\text{ext_to_home}\} + \beta_{ew}\,\Delta\tau_t^{\text{bill}}\cdot\mathbf{1}\{\text{ext_to_work}\} \\ & + \alpha_{(A,B)} + \gamma_t + \varepsilon_{ABt}\end{aligned}$$

Each β is the per-pp dose response within that channel-state. Pair + year FE; SE clustered at work muni.

(5) Piecewise: transition distribution post-Act-32

Pair-years 2012–2019, all 8.6M PA OD-pairs (no distance filter).

	STRICT (drop NA nonres)		*FILLED (NA → work res)*	
Transition	Pair-years	%	Pair-years	%
int_home	1,187,143	39.9	1,604,992	47.0
int_work	1,736,664	58.3	1,755,095	51.4
ext_to_home	19,521	0.7	20,446	0.6
ext_to_work	33,733	1.1	33,930	1.0
Total	2,977,061	100.0	3,414,463	100.0

- Extensive switches are rare (∼1.8% combined). Most pair-year action is intensive within a stable channel.
- Filling shifts pair-years toward home-binding (+7 pp) because imputed nonres = res ≤ res + SD = home stack wherever the imputation kicks in.

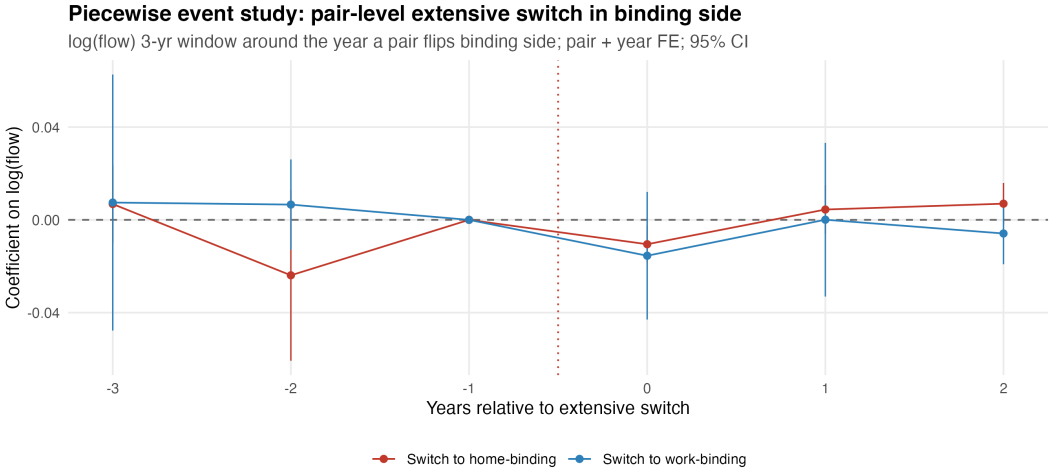
(5) Piecewise: aggregate and decomposed coefficients

log(flow), pair + year FE, SE clustered at work muni.

	STRICT		FILLED	
	Estimate	SE	Estimate	SE
Aggregate $\Delta\tau^{\text{bill}}$	+0.005	0.009	+0.001	0.008
β_{ih} (int_home)	+0.016	0.013	+0.001	0.012
β_{iw} (int_work)	+0.042	0.026	+0.043	0.027
β_{eh} (ext_to_home)	+0.002	0.012	+0.003	0.012
β_{ew} (ext_to_work)	+0.015	0.019	+0.016	0.019
N pair-years	2,735,968		3,129,492	
Pairs	823,244		953,578	

- All four channels null individually. Aggregate $\Delta\tau^{\text{bill}}$ is null. *Direction* is positive across the board, opposite the piecewise prediction.
- Pair-level pair + year FE absorbs almost everything; residual signal of a ∼0.5 pp dose change on a typical flow of 1–3 jobs is too small to detect.
- β_{iw} is the largest at +0.042 (p ≈ 0.11) — wrong-signed positive on intensive work-rate increases. Likely confounded by time-varying muni-A growth that pair FE doesn't

(5) Piecewise event study around extensive switches



76,119 pairs flip to home-binding; 52,570 flip to work-binding. Pre-trends near zero; post-flip drift toward zero is consistent with the null aggregate.

Q3 NONRESIDENT-SIDE: MOTIVATION AND GROUP COMPOSITION

Mirror of resident-side Q3 but treatment = τ_A^{nr} (work muni's nonresident rate); outcome = log(Nonresident workers) in muni A and log(Total Jobs).

- Re-classify munis on nonresident-rate events. Of 88 nonres-Inc munis, **83 had exactly one increase** (94% — multi-event scope is smaller than the resident side). Inc-only: 88; NC: 892; FF / Dec dropped.
- Direct test of the piecewise mechanism: nonresident workers face τ_A^{nr} when the work rate binds; resident workers don't. So the nonres-rate effect should concentrate on nonresident outflows / total jobs, not on resident workers.

(1) Single-event TWFE, nonres-side

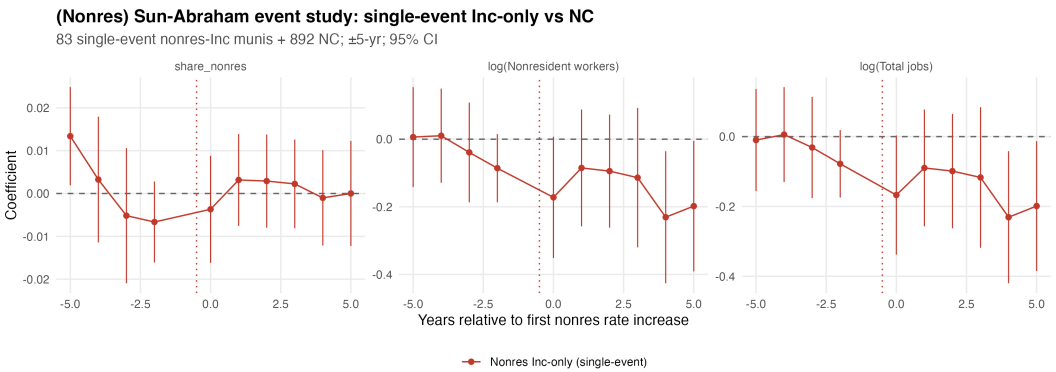
$$Y_{it} = \alpha_i + \gamma_t + \beta\, \tau_{it}^{\text{nr}} + \mathbf{X}_{it}'\boldsymbol{\phi} + \varepsilon_{it}$$

Sample: 892 NC + 83 single-event nonres-Inc munis.

	share_nonres	log(NonRes)	log(Total Jobs)
τ^{nr}	−0.004 (0.012)	−0.289* (0.169)	−0.286* (0.166)
N	9,911	9,911	9,911

Cluster-robust SEs by muni; same controls as resident side.

(1) Single-event ES, nonres-side (Sun-Abraham)



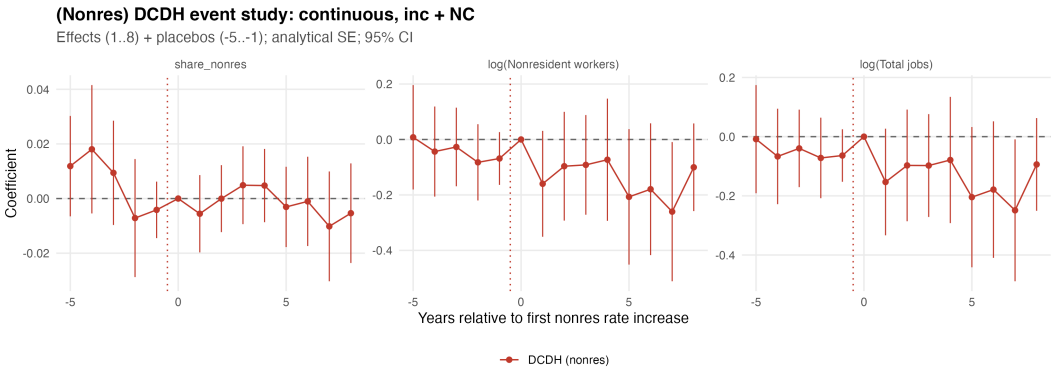
(2) DCDH, NONRES-SIDE: CONTINUOUS, MULTI-EVENT

Same DCDH machinery as resident side; treatment = τ^{nr}.

	share_nonres	log(NonRes)	log(Total Jobs)
Av. total effect	−0.004	−0.298	−0.294
	(−0.028, +0.020)	(−0.663, +0.068)	(−0.646, +0.058)
Switchers	261	261	261
N (cells)	8,807	8,807	8,807

- Point estimates are similar to single-event TWFE (∼−30% on log NonRes and log Total Jobs) but CIs span zero. Bootstrap not yet run.

(2) DCDH ES, NONRES-SIDE



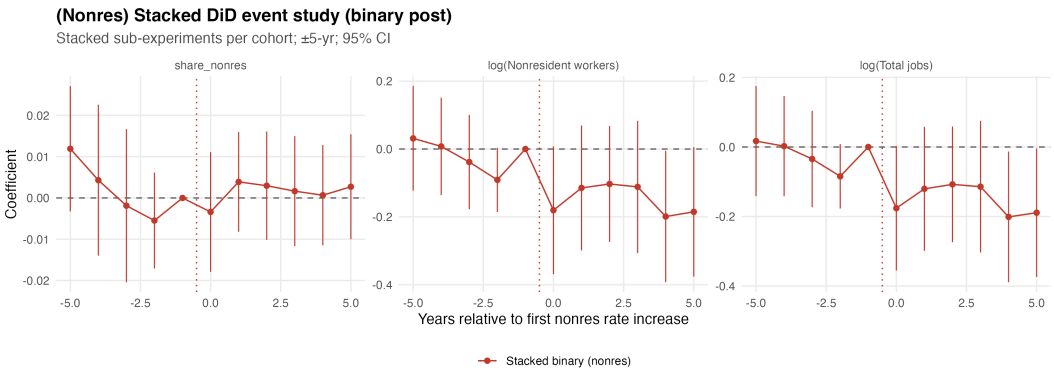
(3) Stacked DiD binary, nonres-side

$$Y_{ist} = \alpha_{is} + \gamma_{ts} + \beta\,(\text{Treat}_i \cdot \text{Post}_{st}) + \mathbf{X}'_{it}\boldsymbol{\phi} + \varepsilon_{ist}$$

18 sub-experiments (cohorts), ±5-yr window, NC-only controls.

	share_nonres	log(NonRes)	log(Total Jobs)
`treat_post`	0.000 (0.007)	−0.129 (0.086)	−0.130 (0.084)
N	76,292	76,292	76,292

(3) Stacked binary ES, nonres-side



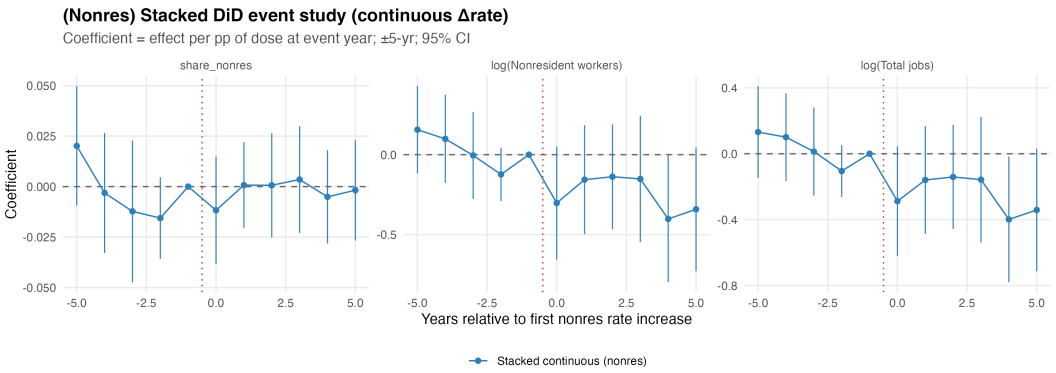
(4) Stacked DiD continuous $\Delta\tau$, nonres-side

$$Y_{ist} = \alpha_{is} + \gamma_{ts} + \beta\,\Delta\tau^{\text{nr}}_{ist} + \mathbf{X}'_{it}\boldsymbol{\phi} + \varepsilon_{ist}$$

	share_nonres	log(NonRes)	log(Total Jobs)
$\Delta\tau^{\text{nr}}$ (per pp)	0.000 (0.013)	−0.261 (0.165)	−0.262 (0.160)
N	61,557	61,557	61,557

- Direction matches resident side (negative); magnitude about 3× larger than resident-side stacked (−0.09).
- p ≈ 0.11. Underpowered: only 88 nonres-Inc munis and small dose variation (most muni nonres rates barely change).
- log(NonRes) and log(Total Jobs) move in tandem — consistent with nonresident workers being a large share of total jobs in many munis.

(4) Stacked continuous ES, nonres-side



Nonres-side estimator triangulation

Estimator	share_nonres	log(NonRes)	log(Total Jobs)
(1) Single-event TWFE	−0.004	−0.289*	−0.286*
(2) DCDH ATE	−0.004	−0.298	−0.294
(3) Stacked binary	+0.000	−0.129	−0.130
(4) Stacked continuous	+0.000	**−0.261**	**−0.262**

- Direction robust: negative on level outcomes across all clean estimators.
- No estimator crosses 5%: nonres-side is underpowered (smaller treated group, smaller dose variation).
- share_nonres null everywhere — mechanically protected because numerator and denominator move together.
- Magnitudes are ∼2–3× resident-side coefficients — nonresident workers are more responsive per pp of tax than residents, in the data and consistent with the piecewise mechanism (residents face the home stack and a larger move cost; nonresidents can switch jobs).

Estimator triangulation: log(Resident) and log(Total Jobs)

Estimator	share_live_in	log(Resident)	log(Total Jobs)	Notes
Q2 TWFE (full)	-0.012^{**}	-0.201^{***}	-0.128^{***}	Pooled, possibly negative-weighted
Single-event TWFE	-0.004	-0.162^{***}	-0.144^{***}	Drops 22 multi-event Inc munis
DCDH ATE	$+0.002$	-0.049	-0.043	Null, wide CI; underpowered
Stacked binary	-0.002	-0.000	$+0.008$	Dose-information lost
Stacked continuous	-0.005	$\mathbf{-0.091}^{**}$	$\mathbf{-0.072}^{**}$	**Recommended headline**

- Direction is robust: *negative* on level outcomes across every clean estimator.
- Magnitude on log outcomes ranges from -0.04 (DCDH) to -0.20 (TWFE pooled). Stacked continuous -0.09 sits in the middle and has the cleanest identification properties.
- `share_live_in` is fragile across clean estimators; the level outcomes are the robust headline.

Summary of Findings

1. Higher muni rates reduce `share_live_in` by about 1.3 share-points per 1 pp, building over 7 to 8 years.
2. The effect is enforcement-contingent. The muni rate coefficient is a precise zero before Act 32 and roughly -1.8 share-points after.
3. The response is asymmetric. Increases push residents out, but decreases do not bring them back.
4. Muni and school district rates have opposite signs. Bundling them as a single "total EIT" rate hides both channels.

Next Steps

- Pair-level piecewise test on OD flows.
- Welfare: combine behavioral elasticity with revenue elasticity to bound efficiency cost of local income tax.

Thank You

Questions?

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github.com/MatthewHockert/local-income-tax-migration